Technical introduction to PSOC™ Edge E84 features

Chapter 1: Introduction and prerequisites

About this document

Scope and purpose

This document is the training manual for the technical introduction to PSOC™ Edge E84 features. This manual helps in getting started with different applications leveraging key features for PSOC™ Edge, using ModusToolbox™ software.

The training is divided into multiple chapters covering topics like audio, graphics, machine learning, among others.

Chapter 1 discusses the prerequisites for the training.

Intended audience

The document is intended for design engineers, technicians, and developers of electronic systems.

Table of contents

About this document 1
Table of contents 2
1 Introduction 3
2 Prerequisites 4
3 Required development tools 5
Revision history 6
Disclaimer 7

1 Introduction

PSOC™ Edge E84 series of Arm® Cortex®-M MCUs feature high-performance, low-power, secured MCUs with advanced machine learning (ML) acceleration for next generation AI/ML applications.

The PSOC™ Edge E84 MCUs feature an Arm® Cortex®-M55 core with Helium™ DSP support, combined with the Ethos™-U55 NPU. They also include a low-power Cortex®-M33 core paired with Infineon's ultra-low-power NNLite hardware accelerator, along with advanced HMI capabilities such as graphics.

This training is intended to provide a brief introduction to some of the key features of PSOC™ Edge through easy-to-use but illustrative hands-on labs.

The training is available in different chapters for easy navigation:

- Chapter 1: Introduction and prerequisites
- Chapter 2: Audio
- Chapter 3: Sensor hub
- Chapter 4: Graphics
- Chapter 5: SMIF (external memories)
- Chapter 6: Low power
- Chapter 7: Machine learning

Each chapter includes a training manual with detailed step-by-step instructions, and source code with solutions.

2 Prerequisites

Install all required software and hardware listed in the Required development tools section.

Note that this class will not cover the basics of ModusToolbox™ and PSOC™ Edge. Before beginning the training, it is recommended to complete an introductory course or a getting started guide to familiarize yourself with ModusToolbox™ and the PSOC™ Edge architecture.

- For an introduction to PSOC™, including a getting started guide to ModusToolbox™, visit PSOC™ Developer Journey
- For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, visit PSOC™ Edge E84 training collection
- To start a first application for PSOC™ Edge using Eclipse IDE for ModusToolbox™, visit Getting started with PSOC™ Edge E8 MCU on ModusToolbox™ software

3 Required development tools

- ModusToolbox™ software v3.6 or later
  - Recommended installation via ModusToolbox™ Setup tool
- Eclipse IDE for ModusToolbox™ v2025.4.0 or later
  - Recommended installation via ModusToolbox™ Setup tool
- Edge Protect Security Suite v1.6 or later
  - Installed by ModusToolbox™ Setup tool as a dependency to ModusToolbox™ v3.6
- ModusToolbox™ Programming Tools v1.5.0 or later
  - Installed by ModusToolbox™ Setup tool as a dependency to ModusToolbox™ v3.6
- Board support package (BSP)
  - KIT_PSE84_EVAL_EPC2: v1.0.0 or later (available with ModusToolbox™ v3.6)
- Serial terminal emulator
  - Eclipse IDE for ModusToolbox™ includes a serial terminal. Other options include Tera Term, Putty, and more
- PSOC™ Edge E84 evaluation kit (KIT_PSE84_EVAL)

The following tools are utilized for audio labs:

- LLVM for Arm® v19.1.5 or later if recommended by ModusToolbox™
  - Optionally, to use LLVM without modifying Makefile: Set the environment variable **CY_COMPILER_LLVM_ARM_DIR** to point to the installation directory
- Create an account for DEEPCRAFT™ Voice Assistant
- Optionally, to observe DEEPCRAFT™ Audio Enhancement tool: DEEPCRAFT™ Audio Enhancement Tech Pack v1.0.0.90 or later
  - Recommended installation via ModusToolbox™ Setup tool

The following tools are utilized for graphics labs:

- Optionally, to create custom LVGL UI: EEZ-Studio v0.24.0 or later
- Optionally, to create custom LVGL UI: Python 3
- Install “pypng” and “lz4” Python modules:
  - `python -m pip install pypng==0.20220715.0`
  - `python -m pip install lz4`
- Optionally, to create custom LVGL UI: Pngquant v2.17.0 or later
  - Download binaries, unzip, and add folder to path environment variable

The following tools are utilized for machine learning labs:

- DEEPCRAFT™ Studio v5.6.3681 or later
  - Create an Imagimob account to train models

Revision history

Document revision	Date	Description of changes
**	2025-01-15	Initial release.

Trademarks

All referenced product or service names and trademarks are the property of their respective owners.

PSOC™, formerly known as PSoC™, is a trademark of Infineon Technologies. Any references to PSoC™ in this document or others shall be deemed to refer to PSOC™.

Edition 2026-01-15

Published by

Infineon Technologies AG

81726 Munich, Germany

© 2026 Infineon Technologies AG. All Rights Reserved.

Do you have a question about this document?

Email:

erratum@infineon.com

Important Notice

Products which may also include samples and may be comprised of hardware or software or both ("Product(s)") are sold or provided and delivered by Infineon Technologies AG and its affiliates ("Infineon") subject to the terms and conditions of the frame supply contract or other written agreement(s) executed by a customer and Infineon or, in the absence of the foregoing, the applicable Sales Conditions of Infineon. General terms and conditions of a customer or deviations from applicable Sales Conditions of Infineon shall only be binding for Infineon if and to the extent Infineon has given its express written consent.

For the avoidance of doubt, Infineon disclaims all warranties of non-infringement of third-party rights and implied warranties such as warranties of fitness for a specific use/purpose or merchantability.

Infineon shall not be responsible for any information with respect to samples, the application or customer's specific use of any Product or for any examples or typical values given in this document.

The data contained in this document is exclusively intended for technically qualified and skilled customer representatives. It is the responsibility of the customer to evaluate the suitability of the Product for the intended application and the customer's specific use and to verify all relevant technical data contained in this document in the intended application and the customer's specific use. The customer is responsible for properly designing, programming, and testing the functionality and safety of the intended application, as well as complying with any legal requirements related to its use.

Unless otherwise explicitly approved by Infineon, Products may not be used in any application where a failure of the Products or any consequences of the use thereof can reasonably be expected to result in personal injury. However, the foregoing shall not prevent the customer from using any Product in such fields of use that Infineon has explicitly designed and sold it for, provided that the overall responsibility for the application lies with the customer.

Infineon expressly reserves the right to use its content for commercial text and data mining (TDM) according to applicable laws, e.g. Section 44b of the German Copyright Act (UrhG).If the Product includes security features:

Because no computing device can be absolutely secure, and despite security measures implemented in the Product, Infineon does not guarantee that the Product will be free from intrusion, data theft or loss, or other breaches ("Security Breaches"), and Infineon shall have no liability arising out of any Security Breaches.

If this document includes or references software:

The software is owned by Infineon under the intellectual property laws and treaties of the United States, Germany, and other countries worldwide. All rights reserved. Therefore, you may use the software only as provided in the software license agreement accompanying the software.

If no software license agreement applies, Infineon hereby grants you a personal, non-exclusive, non-transferable license (without the right to sublicense) under its intellectual property rights in the software (a) for software provided in source code form, to modify and reproduce the software solely for use with Infineon hardware products, only internally within your organization, and (b) to distribute the software in binary code form externally to end users, solely for use on Infineon hardware products. Any other use, reproduction, modification, translation, or compilation of the software is prohibited. For further information on the Product, technology, delivery terms and conditions, and prices, please contact your nearest Infineon office or visit https://www.infineon.com